

Abu Dhabi Real Estate Data Processing Pipeline

DB1 Standardization & Enrichment

Technical Documentation

Prepared for internal data operations and DB1 Ingestion Workflows.

Date: 18 June 2026

Source: ADREC (Abu Dhabi Real Estate Centre)

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1. Executive Summary

This document describes the end-to-end data processing pipeline implemented in **pipeline.ipynb** for Abu Dhabi property transaction records sourced from ADREC. The pipeline ingests raw transaction data, applies deduplication and business-rule transformations, standardizes records into the DB1 schema, enriches location and project coordinates, and assigns non-RERA project indexes for downstream analytics and ingestion.

Key outcomes: ~109,388 deduplicated transaction records transformed into 63 DB1 columns; 132 unique districts geocoded via ArcGIS; 367 unique projects geocoded via Google Maps (362 successful); non-RERA indexes assigned from *nr101* onward.

2. Document Purpose & Scope

Purpose: Provide a complete technical reference for developers, analysts, and data engineers responsible for maintaining, auditing, or extending the Abu Dhabi pipeline.

In scope:

- Raw CSV ingestion and duplicate removal
- Property type and project type standardization
- DB1 column mapping and export
- District-level geocoding (ArcGIS)
- Project-level geocoding (Google Maps / Selenium)
- Non-RERA index generation

Out of scope:

- RERA-indexed project matching
- LLM-based field extraction
- Buyer/seller identity enrichment

3. Pipeline Overview

The pipeline follows a sequential ETL architecture with optional enrichment stages. Each stage produces an intermediate CSV that can be validated independently.

Stage	Input	Output	Tool / Method
1	Abu_Dhabi.csv	In-memory DataFrame	pandas read_csv
2	DataFrame	Deduped DataFrame	drop_duplicates()
3	DataFrame	Transformed DataFrame	transform_abudhabi_transactions()
4	Transformed data	Abu_Dhabi_Cleaned.csv	DB1 column selection
5	Cleaned CSV	geocoded_location_output.csv	ArcGIS geocoder
6	Abu_Dhabi_DB1.csv	Abu_Dhabi_DB1_Project_Geocoded.csv	Google Maps + Selenium

Stage	Input	Output	Tool / Method
7	Abu_Dhabi_DB1.csv	Abu_Dhabi_DB1_Index_Filled.csv	Index assignment logic

Workflow diagram (logical flow):

Raw CSV → Dedupe → Map & Transform → DB1 Export → Location Geocode → Project Geocode → Index Fill → Final DB1 Dataset

4. Input Data Specification

Source file: Abu_Dhabi.csv
Data provider: ADREC (Abu Dhabi Real Estate Centre)
Original record count: 115,514 rows
After deduplication: 109,388 rows
Source columns: 14

#	Source Column	Description
1	Asset Class	High-level asset category (residential, commercial, etc.)
2	Property Type	Detailed property classification (apartment, villa, plot, etc.)
3	Sale Application Date	Transaction date in ISO format (YYYY-MM-DD)
4	Property Sold Area (SQM)	Net sold area in square metres
5	Land Plot Ground Area (SQM)	Land/plot ground area in square metres
6	Property Layout	Unit layout (studio, 1 bed, 2 beds, unclassified, etc.)
7	District	District / location name
8	Community	Community / micro-market identifier
9	Project Name	Development or project name
10	Property Sale Price (AED)	Transaction price in AED
11	Property Sold Share	Fractional share sold (typically 1.0)
12	Rate (AED per SQM)	Derived price per square metre
13	Sale Application Type	off-plan or ready
14	Sale Sequence	primary or secondary sale

5. Processing Stages

5.1 Read Raw Data

Loads Abu_Dhabi.csv using pandas. Displays sample records for visual validation.

5.2 Drop Duplicates

Removes exact duplicate rows using df.drop_duplicates(), reducing records from 115,514 to 109,388 (6,126 duplicates removed).

5.3 Property Type Mapping

Maps 43 raw property types to standardized DB1 categories: Flat, Villa, Office, Shop, Plot, and Others using PROPERTY_TYPE_MAP.

5.4 Project Type Mapping

Maps Asset Class values to standardized project types (Residential, Commercial, Industrial, etc.) using PROJECT_TYPE_MAP.

5.5 Standard Transformation

Core function transform_abudhabi_transactions() applies date parsing, field renaming, derived fields (year, quarter), fixed metadata, and null defaults for unavailable DB1 fields.

5.6 DB1 Column Selection

Selects and orders 63 standard DB1 columns. Missing columns are created with null values. Output saved as Abu_Dhabi_Cleaned.csv.

5.7 Location Geocoding

Geocodes 132 unique district names using ArcGIS API. Address format: '{District}, Abu Dhabi, United Arab Emirates'. Results written to location_latitude and location_longitude.

5.8 Project Geocoding

Geocodes 367 unique projects using Google Maps via Selenium WebDriver. Search format: '{Project Name}, Abu Dhabi, UAE'. 362 projects successfully geocoded; 5 failed.

5.9 Non-RERA Index Assignment

Assigns unique project indexes in format nr101, nr102, ... for records without existing indexes. Same location + project combination reuses the same index.

6. Field Mapping & Transformations

Source Field	DB1 Target Field	Transformation Rule
Project Name	project_name	Direct copy
District	location_name	Direct copy
District	city_name	Fixed to 'Abu Dhabi'
Community	sub_locality, micro_market	Direct copy
Property Sale Price (AED)	agreement_price	Numeric conversion
Property Sold Area (SQM)	net_carpet_area_sq_m	Numeric conversion
Sale Application Date	transaction_date	Parse YYYY-MM-DD → format DD-MM-YYYY
Sale Application Date	date_of_agreement_execution	Same as transaction_date
Sale Application Date	year, quarter	Derived (e.g. Q2-2026)
Property Type	property_type_raw	Lowercase + strip
Property Type	property_type	Mapped via PROPERTY_TYPE_MAP
Property Layout	property_layout	Direct copy (original value retained)
Property Layout	unit_configuration	Lowercase + strip (text retained)
Asset Class	project_type	Mapped via PROJECT_TYPE_MAP
Sale Application Type	furnishing_status	Direct copy (off-plan / ready)
Sale Sequence	sale_type	Direct copy (primary / secondary)
Property Type + Asset Class	property_description	Concatenated with ' '
—	transaction_category	Fixed: 'Sale'
—	country_name	Fixed: 'UAE'
—	state_name	Fixed: 'Abu Dhabi'
—	data_source	Fixed: 'Adrec'
—	source_accessibility	Fixed: 'Download'
—	source_accessibility_way	Fixed: 'Easy'

7. Standardization Rules

7.1 Property Type Mapping (PROPERTY_TYPE_MAP)

Raw Value (sample)	Standardized Value
apartment, duplex, residential complex	Flat
villa, townhouse / attached villa, palace	Villa

Raw Value (sample)	Standardized Value
office, office complex	Office
retail, mall / market / retail center	Shop
plot for *, other * plot	Plot
clinic, factory, hotel, mosque, other, etc.	Others
Unmapped values	Others (default)

7.2 Project Type Mapping (PROJECT_TYPE_MAP)

Raw Asset Class	Standardized project_type
residential	Residential
commercial	Commercial
agricultural	Agricultural
educational	Educational
healthcare	Healthcare
industrial & storage	Industrial
infrastructural	Infrastructure
recreational	Recreational
religious	Religious
other	Others

7.3 Date Handling

Source dates are in ISO 8601 format (YYYY-MM-DD). The pipeline parses using format='%Y-%m-%d' and exports in DD-MM-YYYY format. Quarter is derived as Q{1-4}-{YYYY} (e.g. Q2-2026).

8. Geocoding Enrichment

8.1 District Geocoding (ArcGIS)

Parameter	Value
Geocoder	geopy.geocoders.ArcGIS
Input column	location_name (District)
Search template	{District}, Abu Dhabi, United Arab Emirates
Output columns	location_latitude, location_longitude
Unique locations	132
Rate limiting	1 second sleep between requests
Output file	geocoded_location_output.csv

8.2 Project Geocoding (Google Maps)

Parameter	Value
Method	Selenium WebDriver + Google Maps search URL
Input column	project_name
Search template	{Project Name}, Abu Dhabi, UAE
Output columns	project_latitude, project_longitude
Unique projects	367
Success rate	362 OK / 5 failed
Output file	Abu_Dhabi_DB1_Project_Geocoded.csv

9. Index Assignment

Records without an existing **index** value receive a non-RERA identifier starting from **nr101**. The assignment logic:

- Groups records by (location_name, project_name) combination
- Reuses existing index if the same combination already has one
- Increments sequentially (nr101, nr102, nr103, ...) for new combinations
- Skips rows where project_name is null or blank
- Output saved as Abu_Dhabi_DB1_Index_Filled.csv

10. Output Deliverables

File Name	Description	Records
Abu_Dhabi_Cleaned.csv	DB1-standardized transactions (63 columns)	~109,388
geocoded_location_output.csv	Cleaned data + district lat/long	~109,388
Abu_Dhabi_DB1_Project_Geocoded.csv	DB1 data + project coordinates	~109,388
Abu_Dhabi_DB1_Index_Filled.csv	Final dataset with nr-indexes assigned	~109,388

11. Dependencies & Execution

Python libraries required:

Library	Purpose
pandas	Data loading, transformation, export
numpy	Numeric operations and null handling
geopy	ArcGIS district geocoding
selenium	Browser automation for Google Maps
webdriver_manager	ChromeDriver management
tqdm	Progress tracking during geocoding

Execution order in pipeline.ipynb:

- Cell 0–2: Import libraries and load Abu_Dhabi.csv
- Cell 4: Remove duplicate rows
- Cell 7, 9: Define PROPERTY_TYPE_MAP and PROJECT_TYPE_MAP
- Cell 11: Define transform_abudhabi_transactions()
- Cell 12–16: Transform, select DB1 columns, export Abu_Dhabi_Cleaned.csv
- Cell 19: District geocoding (ArcGIS)
- Cell 21: Project geocoding (Google Maps / Selenium)
- Cell 23: Non-RERA index assignment

12. Data Quality & Limitations

Area	Limitation	Impact
Furnishing status	No furnishing field in source; Sale Application Type used as proxy	Sales are off-plan/ready, not furnished/unfurnished
Unit configuration	Non-bedroom layouts (unclassified, commercial size) not mapped	Unit configuration values remain as raw text
DB1 null fields	35+ DB1 fields unavailable in source (buyer, seller, Regulated, etc.)	All in output
Geocoding	API-based; subject to rate limits and address ambiguity	5 projects failed geocoding; some coordinates may be approximate
transaction_type	Listed in DB1_COLUMNS but not explicitly mapped	Columns reported as null unless added manually

13. Appendix: DB1 Output Schema (63 Columns)

#	Column Name
1	project_name
2	location_name
3	registered_document_village_name
4	agreement_price
5	transaction_date
6	net_carpet_area_sq_m
7	sub_registrar_office_code
8	sub_registrar_office_name
9	document_number
10	transaction_type
11	city_name
12	guideline_value
13	property_description
14	floor_number
15	unit_number
16	property_type_raw
17	property_type
18	year
19	balcony_sq_m
20	terrace_sq_m
21	seller_name

#	Column Name
22	buyer_name
23	transaction_category
24	internal_document_number
25	micr_number
26	bank_type
27	party_code
28	date_of_agreement_execution
29	stamp_duty_paid
30	registration_fee
31	project_latitude
32	project_longitude
33	location_latitude
34	location_longitude
35	quarter
36	unit_configuration
37	property_layout
38	buyer_pincode
39	buyer_locality
40	buyer_district
41	buyer_state
42	is_llm_processed
43	is_manual_processed
44	tower_name
45	is_duplicate
46	sale_type
47	project_type
48	country_name
49	state_name
50	micro_market
51	sub_locality

#	Column Name
52	pincode
53	parking_count
54	facing_direction
55	view_type
56	furnishing_status
57	condition_status
58	source_accessibility
59	source_accessibility_way
60	sourcing_cost
61	sourcing_time
62	data_type
63	data_source

End of Document — Abu Dhabi Real Estate Data Processing Pipeline